PHY 211

Lecture 9 Notes

By

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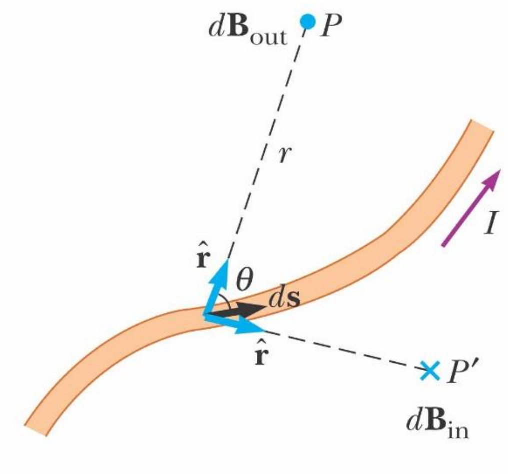
Chapter 8
Source of the Magnetic Fields

➢ The magnetic field produced by the current "Biot-Savart law":

- Biot and Savart conducted experiments on the force exerted by an electric current on a nearby magnet
- They arrived at a mathematical expression that gives the magnetic field at some point in space due to a current
- That expression is based on the following experimental observations for the magnetic field $d\vec{B}$ at a point P due to a length element $d\vec{s}$ of a wire carrying a steady current I:

1. The vector $d\vec{B}$ is perpendicular both to $d\vec{s}$ (which points in the direction of the current) and to the unit vector $\hat{r}$ directed from $d\vec{s}$ toward P.
2. The magnitude of $d\vec{B}$ is inversely proportional to r^2, where r is the distance from $d\vec{s}$ to P.
3. The magnitude of $d\vec{B}$ is proportional to the current I and to the magnitude ds of the length element $d\vec{s}$.



4. The magnitude of $d\vec{B}$ is proportional to $\sin\theta$, where θ is the angle between the vectors $d\vec{s}$ and $\hat{r}$.

- These observations are summarized in the mathematical expression known today as the **Biot–Savart law**:

$$\boxed{d\vec{B} = \frac{\mu_o}{4\pi}\frac{I\, d\vec{s} \times \hat{r}}{r^2}} \rightarrow \boxed{dB = \frac{\mu_o}{4\pi}\frac{I\, ds\, \sin\theta}{r^2}}$$

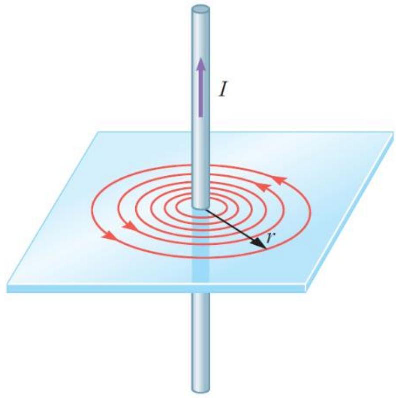
μ_o: constant called Magnetic permeability of free space.

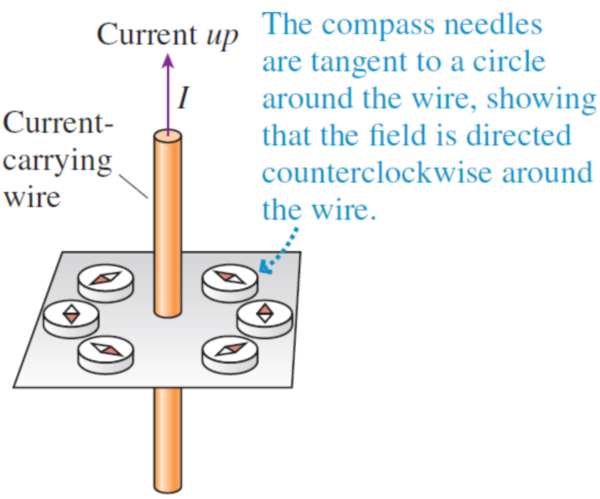
$$\mu_o = 4\pi \times 10^{-7}\ \frac{\mathrm{T \cdot m}}{\mathrm{A}}$$

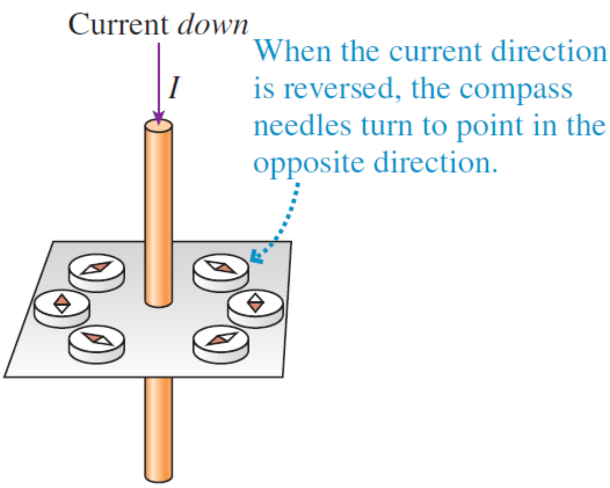
➢ Applications of Biot-Savart's law:

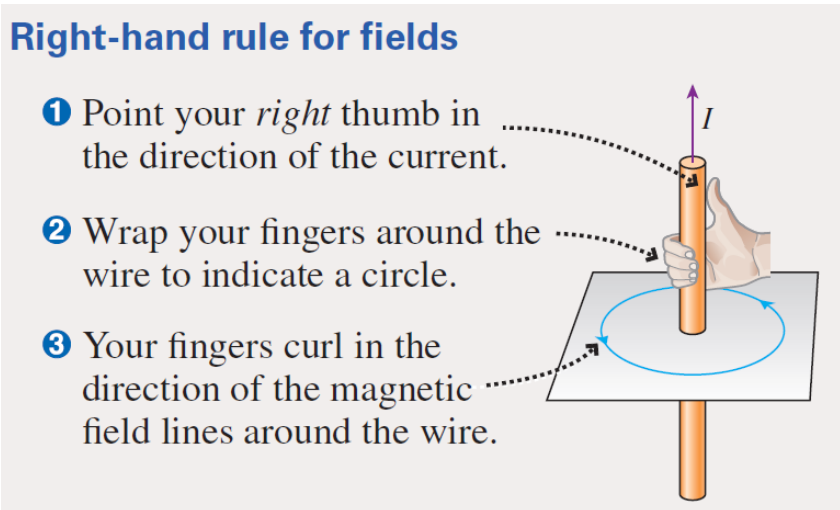
1- The magnetic field of a thin straight, current-carrying wire:

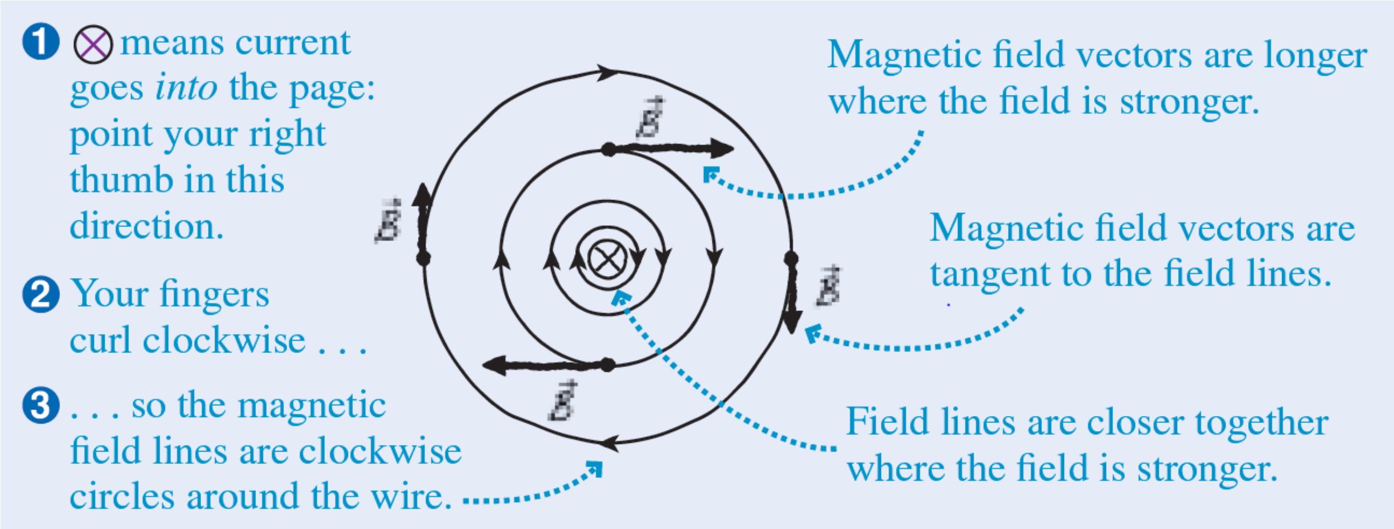
$$\boxed{B = \frac{\mu_o I}{2\pi r}} \rightarrow B \propto I \;\&\; B \propto \frac{1}{r}$$

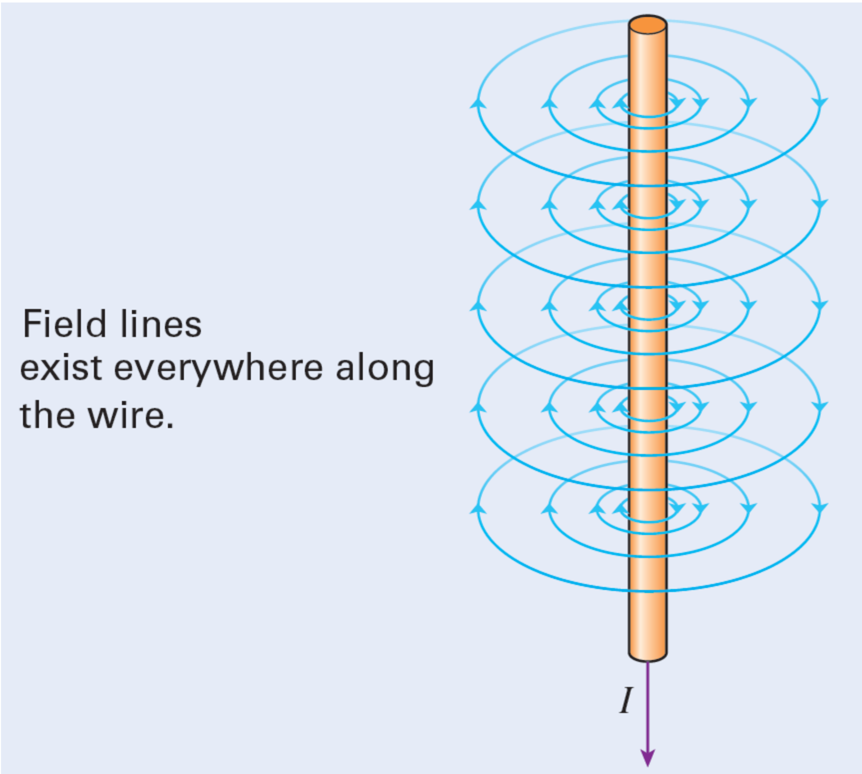




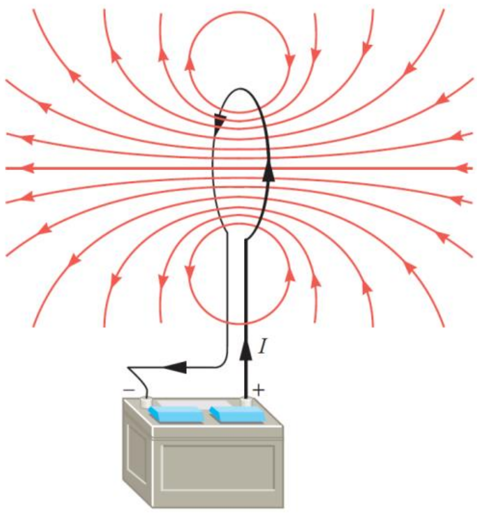


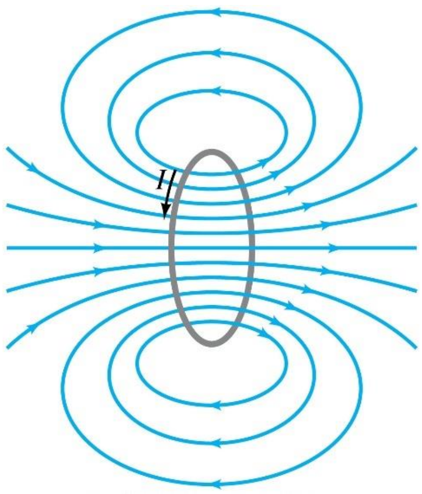


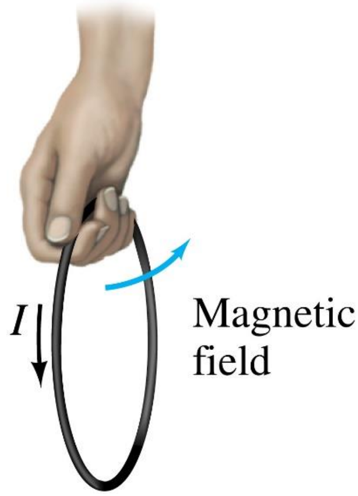


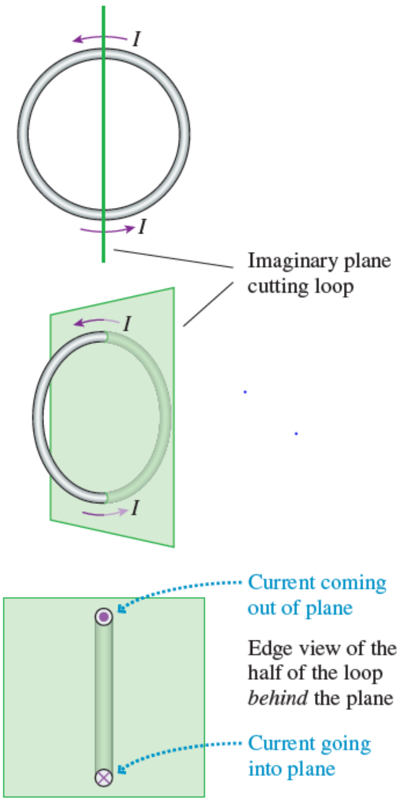


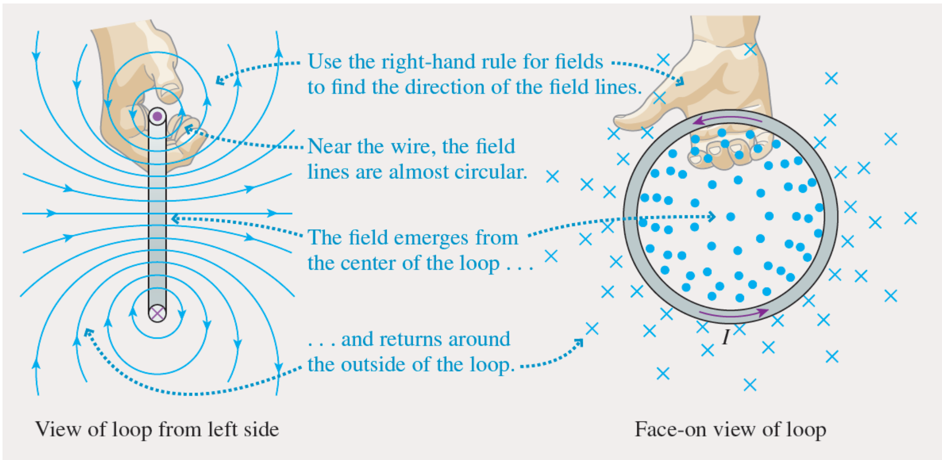
2- The magnetic field of a current loop:

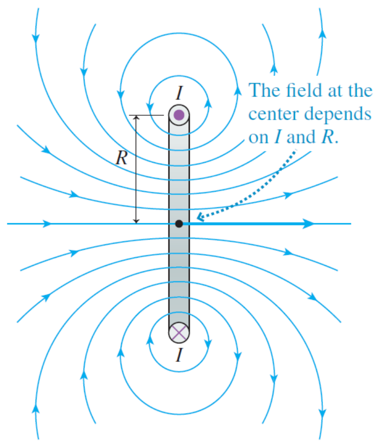












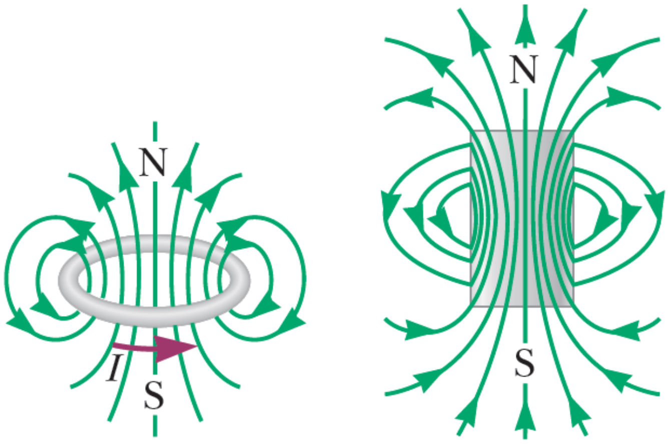
- The magnetic field at the center of a current loop of radius R is given by:

$$\boxed{B = \frac{\mu_o I}{2R}}$$

- The magnetic field at the center of a thin coil of N turns of loops:

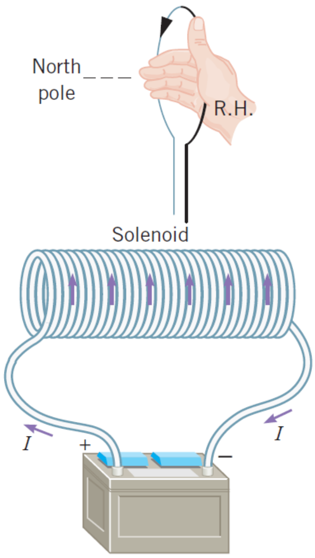
$$\boxed{B = \frac{\mu_o N I}{2R}}$$

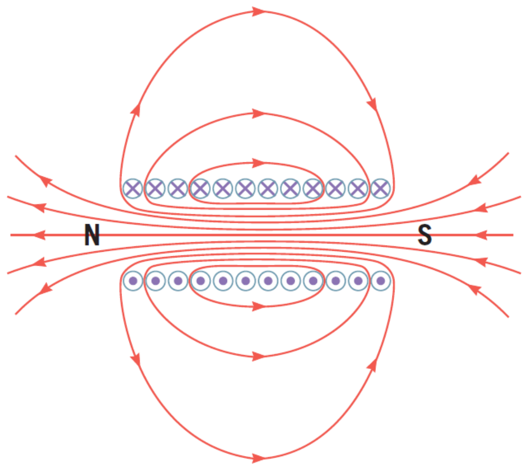
- The Magnetic field lines surrounding a current loop is similar to the magnetic field lines surrounding a **short bar magnet**.



3- The magnetic field of a Solenoid:

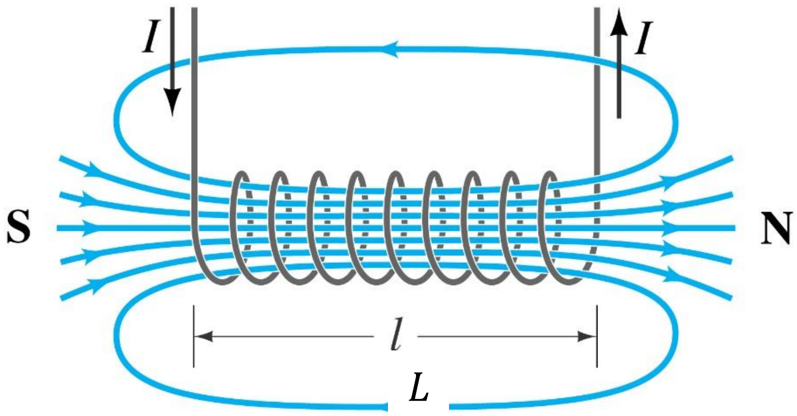
- A **solenoid** is a long wire wound in the form of a helix.





A solenoid and a cross-sectional view of it, showing the magnetic field lines and the north and south poles.

- The Magnetic field lines of a solenoid is similar to the magnetic field lines surrounding a **long bar magnet**.



- The magnitude of the magnetic field in the interior of a long solenoid of N turns and length L is

$$\boxed{\boldsymbol{B = \mu_o n I}}$$

$$\boxed{\boldsymbol{n = \frac{N}{L}}} \textbf{ is the number of turns per unit length}$$

- The magnetic field inside the solenoid is uniform (constant value and directed parallel to the axis)

- **Important note:**
- The length of the wire "d" forming the Solenoid is given by:

$$\boxed{\textbf{Length of wire} \quad \boldsymbol{d = N \times 2\pi r}}$$

- **Example 1:** A research student is preparing a solenoid to be used in studying the effect of a magnetic field on a certain type of bacteria. If the required magnetic field inside the solenoid is $0.002\ \mathrm{T}$, the length of the solenoid is $0.25\ \mathrm{m}$ and the current passing through it is $2.5\ \mathrm{A}$. What would be the length of the wire required to construct this solenoid knowing that its radius is $2\ \mathrm{cm}$?

Solution

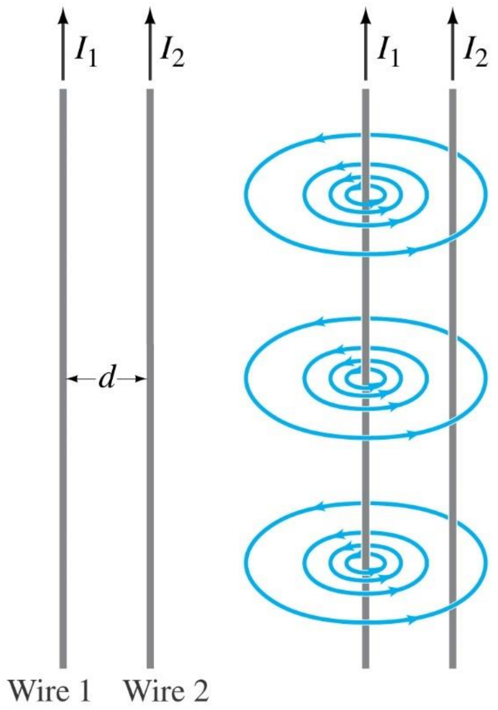
$$B = \mu_o n I = \mu_o \frac{N}{L} I$$

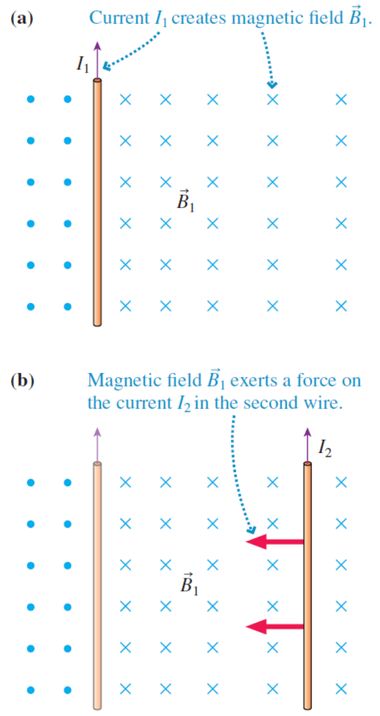
$$\therefore N = \frac{LB}{\mu_o I} = \frac{0.25 \times 0.002}{(4\pi \times 10^{-7}) \times 2.5} = 159$$

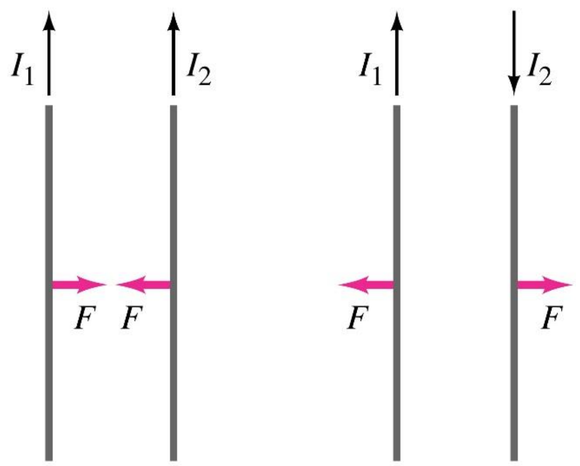
$$Length\ of\ wire = N \times 2\pi r = 159 \times 2\pi \times 0.02 = 20\ \mathrm{m}$$

➢ Forces between currents:

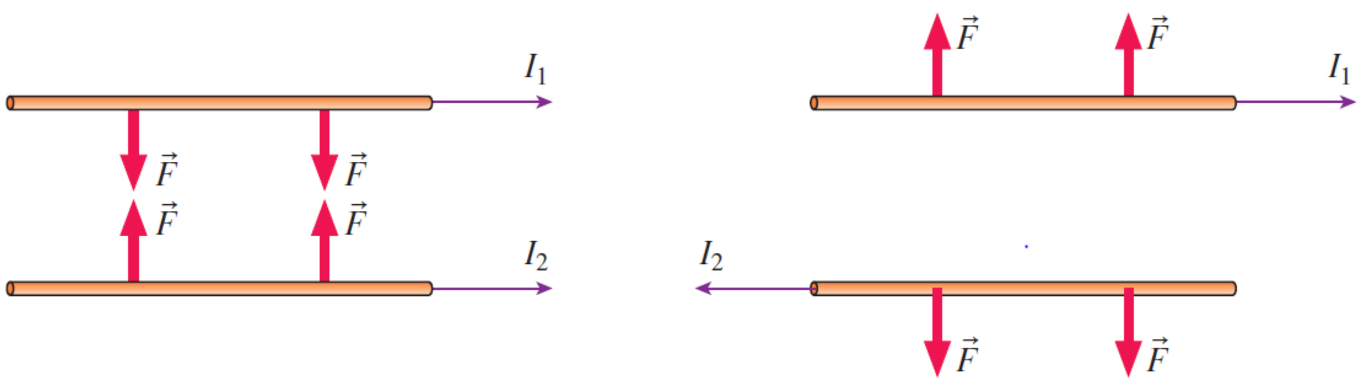
1- Magnetic force between two parallel current-carrying wires:



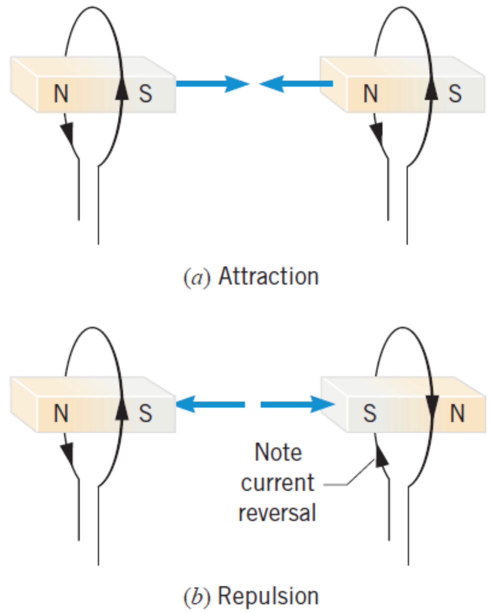




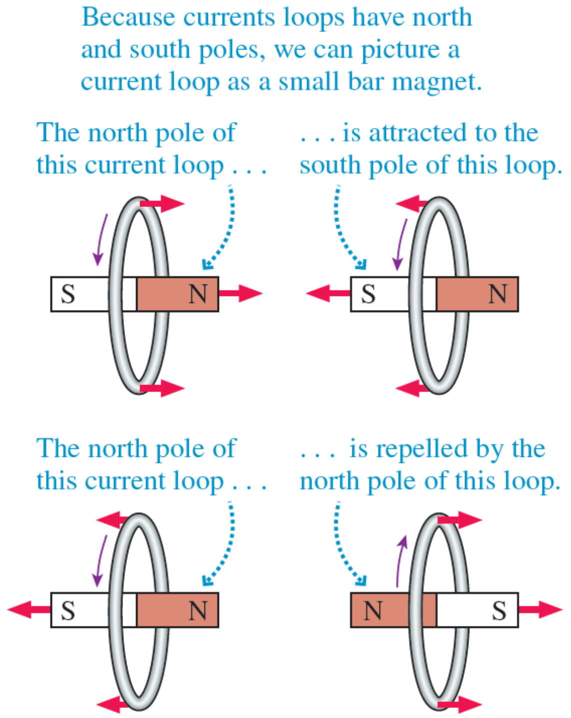
(a) Currents in the same direction attract. **(b)** Currents in opposite directions repel.



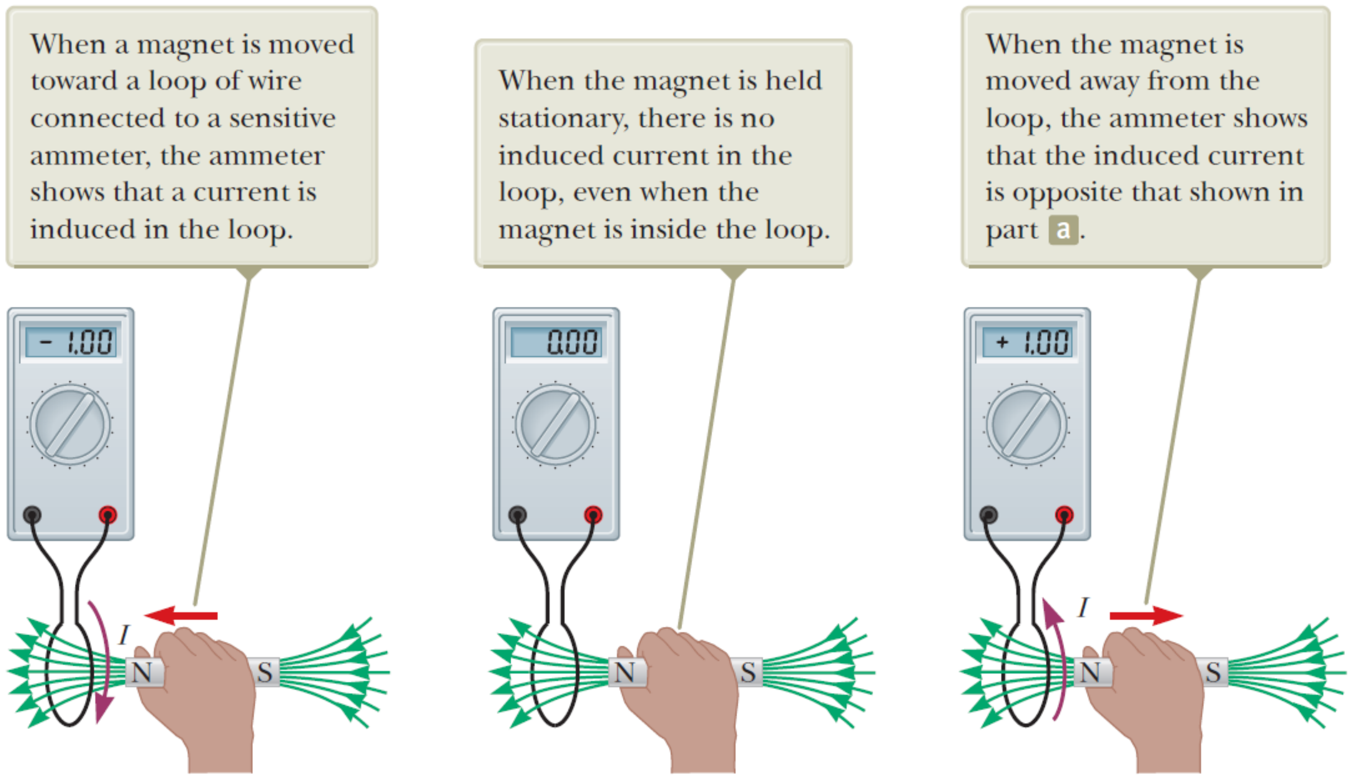
2- Magnetic force between two current loops:



(a) The two current loops attract each other if the directions of the currents are the same and (b) repel each other if the directions are opposite.



➢ Lenz's law:



- **Lenz's law:** There is an induced current in a closed, conducting loop if and only if the magnetic flux through the loop is changing. The direction of the induced current is such that the induced magnetic field opposes the change in the flux.